AI and Health – Introduction

BMED365: What is AI, History, and AI/ML/DL Concepts

BMED365

Spring 2026

Overview

1. **What is Artificial Intelligence?**
   - Definition of AI
   - Narrow vs. General AI
   - The Turing Test
   - AI in everyday life and healthcare
2. **AI in Healthcare: A Historical Journey**
   - Timeline: 1950s to today
   - Key milestones and systems
   - Lessons from history
   - Future perspectives
3. **AI, ML, and DL: Understanding the Terminology**
   - Hierarchical relationship
   - Machine Learning fundamentals
   - Deep Learning and neural networks
   - Decision guide: Which approach to choose?

What is Artificial Intelligence?

Definition

Artificial Intelligence (AI) are systems that can:

- Solve tasks that normally require human intelligence
- Learn from experience
- Adapt to new situations
- Reason and make decisions

Examples of AI:

- Google Maps route finding
- Automatic text translation
- Facial recognition
- Spam filters

NOT AI:

- Simple calculator
- Thermostat
- Database storage

Narrow vs. General AI

Narrow AI (ANI):

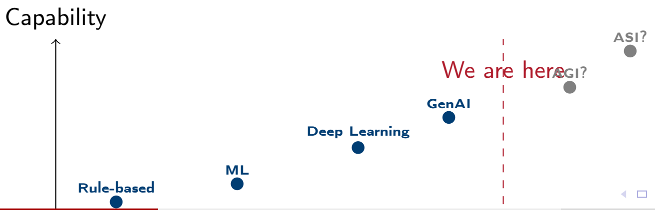
- Designed for specific tasks
- All current AI is narrow AI
- Examples:
  - Chess AI
  - Image recognition
  - Language translation

General AI (AGI):

- Can solve any intellectual task
- Human-like intelligence
- Doesn't exist yet
- (Maybe never will?)

Superintelligence (ASI):

- Surpasses human intelligence
- Science fiction (for now!)



The Turing Test: Can Machines Think?

Alan Turing (1950)

A person chats with both a human and a machine. If the person cannot tell them apart → The machine is "intelligent".

AI Response:

- Structured, fact-based
- Professional tone
- Consistent

"Based on the symptoms, this could be tension headache. Common triggers include stress, dehydration, and poor sleep."

Human Response:

- Personal anecdotes
- Emotional language
- Less formal

"Oh, headaches can be so annoying! I remember when I had migraines... Have you tried lying in a dark room?"

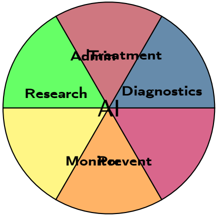
Relevance for Healthcare

The Turing Test raises questions about AI's role in patient communication and whether AI can truly understand human needs.

AI in Healthcare Today

Healthcare Areas:

- **Diagnostics:** Image analysis (X-ray, MRI)
- **Treatment:** Robotic surgery, medication optimization
- **Administration:** Medical records, scheduling
- **Research:** Drug development, genomics
- **Monitoring:** Wearables, remote monitoring
- **Prevention:** Risk prediction, health apps



Strengths of Today's AI

Processes large amounts of data quickly, finds patterns humans overlook, consistent performance 24/7, can be specialized for specific tasks.

AI in Healthcare: Historical Timeline

1950-1980: Foundation

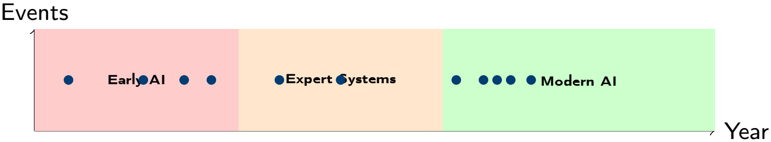
- 1950: Turing Test
- 1966: ELIZA (therapy chatbot)
- 1972: MYCIN (antibiotic system)
- 1976: INTERNIST-I

1980-2010: Expert Systems

- 1986: DXplain launched
- 1995: Robot-assisted surgery

2010-Today: AI Revolution

- 2012: Deep Learning breakthrough
- 2016: AlphaGo defeats Go master
- 2018: FDA approves AI for diabetic retinopathy
- 2020: COVID-19 accelerates AI adoption
- 2023: ChatGPT and generative AI



Key Milestones: Early Systems

1950: Turing Test

- Foundation for assessing AI
- Still relevant today

1966: ELIZA

- First "therapist chatbot"
- Simulated Rogerian psychotherapist
- Showed potential and limitations

1972: MYCIN

- Expert system for antibiotics
- Diagnosed bacterial infections
- Performed as well as experts

1976: INTERNIST-I

- Comprehensive diagnostic system
- Handled thousands of diseases
- Demonstrated complexity

1986: DXplain

- First commercial medical expert system
- Still in use today (updated)
- Shows importance of continuous development

1995: Robot-assisted surgery

- da Vinci system introduced
- Revolutionized precision
- Combined AI with robotics

Modern AI Revolution (2010-Today)

2012: Deep Learning Breakthrough

- CNNs win ImageNet
- Starts modern AI revolution
- Opens advanced medical image analysis

2016: AlphaGo

- Defeats world champion in Go
- Demonstrates strategic thinking
- Inspires complex medical decision-making

2018: FDA Approval

- First autonomous AI system approved
- IDx-DR for diabetic retinopathy
- Milestone for AI as medical device

2020: COVID-19

- AI accelerates vaccine development
- Rapid diagnosis from imaging
- Demonstrates AI's value in crises

2023: Generative AI

- ChatGPT and similar models
- New possibilities for medical writing
- Paradigm shift in AI interaction

Lessons from History

Success Factors:

1. **Specialization first:** Early systems like MYCIN focused on narrow domains
2. **Iterative improvement:** DXplain continuously evolved since 1986
3. **Regulatory approval:** FDA process ensures safety
4. **Crises as catalyst:** COVID-19 accelerated adoption

Challenges and Failures:

1. **Over-optimism:** Many early systems promised more than delivered
2. **User acceptance:** Physicians were skeptical
3. **Integration:** Difficult to integrate into workflows
4. **Responsibility:** Who is responsible when AI makes mistakes?

Key Insight

AI is powerful, but requires thoughtfulness, testing, and gradual implementation. History shows that successful AI systems are those that solve specific problems well and evolve over time.

Future Perspectives: Next 5-10 Years

Expected Developments:

- **Multimodal AI:** Combining text, images, sensors
- **Personalized medicine:** AI tailored to individuals
- **Real-time monitoring:** Continuous health monitoring
- **Robotics:** Advanced surgical and care robots

Expected AI Maturity (2024 → 2030):

- Medical image diagnostics: 7/10 → 9/10
- Drug development: 4/10 → 8/10
- Patient monitoring: 5/10 → 8/10
- Surgical assistance: 3/10 → 7/10

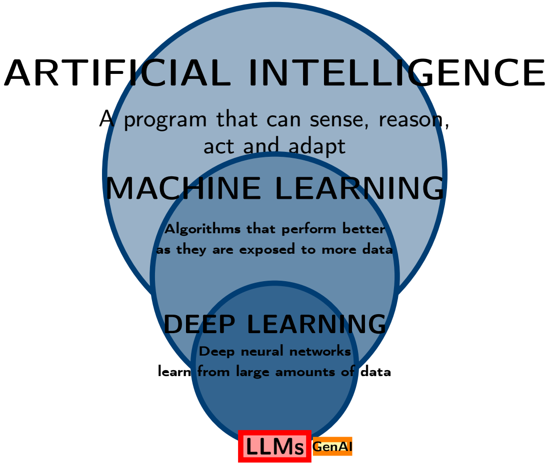
Challenges Ahead:

- **Data quality and bias:** Ensure representative datasets
- **Interoperability:** AI systems that communicate
- **Ethics and regulation:** Balance innovation with safety
- **Education:** Prepare healthcare personnel

The Future

AI will transform all aspects of healthcare, but requires careful implementation and ethical considerations.

AI, ML, and DL: The Hierarchical Relationship



Key Point

Deep Learning $\subseteq$ Machine Learning $\subseteq$ Artificial Intelligence

Machine Learning: From Mystery to Mathematics

The Machine Learning Formula

$$y \approx f(x; \theta)$$

- x = Input (data: table, text, image, audio, ...)
- f = Model (function/algorithm)
- θ = Model parameters (learned from data)
- y = Output (diagnosis, prognosis, grade, type, ...)

Why Mathematics?

- **Clarity:** Precise understanding
- **Communication:** Speak with colleagues
- **Problem Solving:** Debug and improve

Why "$\approx$" not "$=$"?

- Models give approximations
- Always uncertainty and error
- In healthcare: "85% probability of diabetes", not "definitely diabetes"

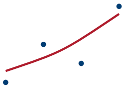
Important

AI results must always be interpreted with caution! The symbol $\approx$ reminds us that predictions are probabilistic, not

Types of Machine Learning

Supervised Learning:

- Data has **labels**
- Learn input→output mapping
- Types: Classification, regression
- Medical: Diagnosis, prognosis

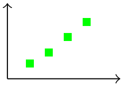


Unsupervised Learning:

- Data **without** labels
- Finds hidden patterns
- Types: Clustering, dimensionality reduction
- Medical: Patient subgroups

Reinforcement Learning:

- Environment with reward/punishment
- Optimize actions
- Medical: Medication dosing optimization



Deep Learning: ML with Neural Networks

Definition

Deep Learning is a subset of machine learning that uses neural networks with many layers ("deep" networks) to learn complex patterns.

Characteristics:

- Neural networks with many layers
- Automatic feature engineering

Deep Learning: The Future of Personalized Medicine

Vision from E. Topol (Nature Medicine, 2019)

Deep learning can revolutionize healthcare by combining **all available data** about a patient into **individualized treatment**.

Input (x) - Multimodal data:

- Genomics (DNA, RNA, proteins)
- Clinical data (lab values, vital signs)
- Lifestyle (activity, sleep, diet)
- Cognitive state
- Medical literature
- Biosensors (real-time data)

Output (y) - Personalized guidance:

- Precise diagnosis
- Tailored treatment
- Optimal timing
- Prognosis

Why Revolutionary?

- Holistic approach
- Individualization
- Continuous improvement
- Prevention focus

Challenges

Comparison: When Do We Use What?

Aspect	Rule-based	Traditional ML	Deep Learning
Data volume	Low	Medium	High
Interpretability	High	Medium	Low
Computing power	Low	Medium	High
Development time	Short	Medium	Long
Accuracy (complex data)	Low	Medium	High

Healthcare Examples:

- **Rule-based:** Medication dosing, allergy alerts
- **Traditional ML:** Risk scores, patient grouping
- **Deep Learning:** Image analysis, language processing

Case Study: X-ray Analysis

- **Rule-based:** 60-70% accuracy
- **Traditional ML:** 75-85% accuracy
- **Deep Learning:** 90-95% accuracy

Decision Guide: Which Approach Should I Choose?

Answer These Questions:

1. **How much data do you have?**
   - Little (< 1000) → Rule-based
   - Moderate (1000-100k) → Traditional ML
   - A lot (> 100k) → Can consider Deep Learning
2. **How important is interpretability?**
   - Critical (medical diagnosis) → Rule-based or interpretable ML
   - Important → Traditional ML
   - Less important → Deep Learning OK

More Questions:

3. **How complex are your data?**
   - Simple (numbers, categories) → Rule-based/Traditional ML
   - Complex (images, text, audio) → Deep Learning
4. **How much time/resources do you have?**
   - Limited → Rule-based

Summary: Key Points

What is AI?

- Systems that solve tasks requiring human intelligence
- Narrow AI (current) vs. General AI (future)
- Turing Test: Can machines think?
- AI in everyday life and healthcare

AI History

- 1950s: Foundation (Turing, ELIZA, MYCIN)
- 1980s-2000s: Expert systems (DXplain, robotics)
- 2010s: Deep Learning revolution
- 2020s: Generative AI and multimodal systems
- Lessons: Specialization, iteration, regulation

AI/ML/DL Concepts

- DL $\subseteq$ ML $\subseteq$ AI (hierarchical)
- ML: $y \approx f(x; \theta)$
- Supervised/Unsupervised/Reinforcement
- Deep Learning: Neural networks with many layers
- Decision guide: Start simple!

Practical Advice

Start simple – build baseline first. **Consider data volume** – more data = more complex methods. **Think interpretability** – critical in medicine. **Resources** – deep learning requires time and computing power.

Thank you! Questions?